

1. Part 1. Linear Regression

- (a). The value θ_0 represents the regression equation's y-intercept. The Linear Regression Model produced a value of $\theta_0 = -1.83$.

θ_1, θ_2 , and θ_3 represent the coefficients which Carbohydrates, Protein and Total Fat are multiplied by, respectively. These values are considered the features' weights or independent variables. The values for the features that the Linear Regression Model produced are:

$$\theta_0 = -1.83, \theta_1 = 3.98, \theta_2 = 4.05, \theta_3 = 9.02.$$

The dependent variable or label in this instance is the Calories. It is the sum of all features multiplied by their respective coefficients, added to the y-intercept.

- (b). With only using the Total Fat feature, the model produced a y intercept $\theta_0 = 151.59$ and the only coefficient $\theta_1 = 15.30$.

These values differ significantly because only one feature is being used to predict an equation that inherently depends on. As a result, the accuracy and precision of the model are significantly reduced.

- (c). The result of training the ML model on all numeric values is as follows: MSE of the train-set: 20.60 MSE of the test-set: 26.79. The MSE of the test-set is higher because the ML model was over-fitted to the attributes train-set. Overfitting occurs when the ML model is trained on the training set's noise in addition to its general trends.

2. Part 2. Logistic Regression

- (a). The Adult income dataset has nine 'object' data-type attributes, and six "integer" attributes. One-hot encoding was performed on all nine 'object' data types, and then aggregation was performed on six of the sets of new attributes. This resulted in a set of 44 independent attributes.

- (b). 44 attributes were too many to train the Logistic Regression model with as many had very little correlation with respect to the target variable. Therefore, sklearn's Recursive Feature Elimination function (RFE) class was implemented with the model to select the top 10 attributes.

The results from the Logistic Regression Model were insufficient, as a low recall value (0.28) drastically affected the model's ability to predict 'Incomes greater than or equal to 50k' (denoted as '1' in the Classification Reports) and resulted in a ROC-AUC value of 0.61. The first attempt to fix this was to train the model two more times, once with 5 attributes, and once with 15 attributes. Both trials had little impact on the model's metrics. This retraining is not shown in the code for the sake of brevity.

```

auc: 0.61
model score: 0.80

```

	precision	recall	f1-score	support
0	0.81	0.97	0.88	7387
1	0.73	0.28	0.40	2382
accuracy			0.80	9769
macro avg	0.77	0.62	0.64	9769
weighted avg	0.79	0.80	0.76	9769

Figure 1: Logistic Regression Model 1 Metrics

Next, the five most positively correlated and the five most negatively correlated attributes were used to train the model. This increased the model's recall to 0.37, and the ROC-AUC to 0.84. This was a significant increase. However, there was still a significant amount of error to address, therefore L1 and L2 models were implemented in the attempt to get better results.

```

auc: 0.84
model score: 0.81

```

	precision	recall	f1-score	support
0	0.83	0.95	0.89	7474
1	0.69	0.37	0.48	2295
accuracy			0.81	9769
macro avg	0.76	0.66	0.69	9769
weighted avg	0.80	0.81	0.79	9769

Figure 2: Logistic Regression Model with 50/50 split for Positive and Negative Correlation

The L2 model provided results similar to the first model, however the L1 model provided the best results yet with an ROC-AUC value of 0.90. All of the values in the classification report increased, with the recall field for classification 1 increasing to 0.58.

```

auc: L1 0.90
model score: 0.85

```

	precision	recall	f1-score	support
0	0.88	0.94	0.91	7446
1	0.75	0.58	0.65	2323
accuracy			0.85	9769
macro avg	0.81	0.76	0.78	9769
weighted avg	0.85	0.85	0.85	9769

Figure 3: L1 Regression Model

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auc: L2 0.60
model score: 0.80

```

	precision	recall	f1-score	support
0	0.81	0.97	0.88	7446
1	0.73	0.27	0.39	2323
accuracy			0.80	9769
macro avg	0.77	0.62	0.64	9769
weighted avg	0.79	0.80	0.77	9769

Figure 4: L2 Regression Model

- (c). K-folds classification was performed on the L1 model in order to improve the best performing model. However, no significant changes occurred. Additionally, this model produced 10 highly precise but fairly accurate models. This affirms that the the L1 model was incredibly consistently with this Model.

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Stratified K-Fold metrics for '1' values with L1 Regression Model:

Average accuracy:
0.85

Average precision:
0.73

Average recall:
0.59

Average f1:
0.65

Average roc_auc:
0.90

```

Figure 5: L1 Regression Model with K-folds Stratification Metrics

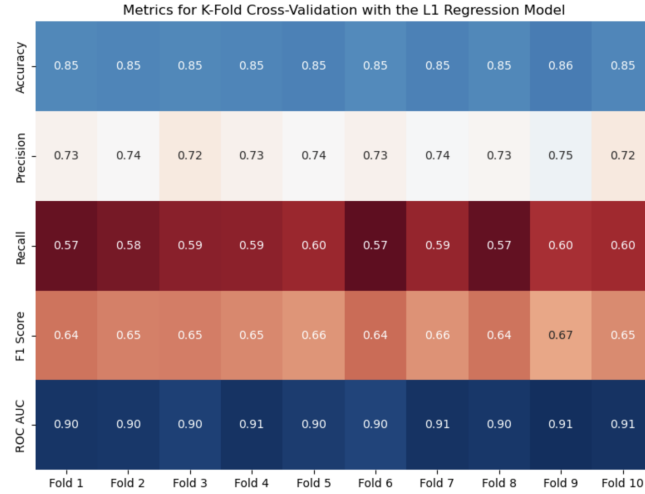


Figure 6: L1 Regression/K-folds Confusion Matrix

In conclusion, the L1 model performed the best across nearly all metrics, with the greatest improvements seen with the ‘recall’ metric. The model did not improve any better when implemented with a K-folds Stratification. The remaining discrepancies are most likely due to non-linear relationships between some of the features and the target value. These issues could likely be solved with a Decision Tree Classifier ML Model.